

DEM 601 Introduction to Demographic Analysis

Lecture 2 handout TOPIC Demographic Indicators and Person-Years-Lived

SESSION NAME
1 October 2026

The shape of the lecture, not a substitute for it. Every heading is a slide, in the order it is shown, with the formulas it carries and the terms it defines.

Opening

WHERE WE ARE From counting to comparing

Why Indicators

COMPARISON The same growth, twice

DEFINITION What an indicator is

$$\text{indicator} = \frac{\text{numerator}}{\text{denominator}} \times k$$

YOUR TURN Name the four parts

Summarising a Distribution

ARITHMETIC The mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

ARITHMETIC The mean, worked

ARITHMETIC What the mean hides

ARITHMETIC The median

An odd number of values the middle one is a real observation

An even number of values the middle falls between two of them

ARITHMETIC The median, worked

ARITHMETIC The median of grouped data

$$M = L_0 + \left(\frac{P/2 - \sum f_0}{f_0} \right) i$$

ARITHMETIC Mean or median?

YOUR TURN Why the median, every time?

Five Words, Five Meanings

COMPARISON Ratio

$$\text{ratio} = \frac{a}{b} \times k$$

COMPARISON Proportion

$$\text{proportion} = \frac{a}{a+b}$$

COMPARISON Percentage

$$\text{percentage} = \frac{a}{a+b} \times 100$$

COMPARISON One question tells them apart

RATE A rate needs time

$$\text{rate} = \frac{e}{PY} \times k$$

An event must occur something happens, not something is

A group must be able to experience it the denominator has to be at risk

A period must be stated a rate without time is not a rate

RATE Crude, specific, standardised

Crude everybody in the denominator

Specific only the people at risk

Standardised a common age structure imposed

RATE Is it a true rate?

$$CBR = \frac{\text{births in the year}}{\text{mid-year population}} \times 1000$$

$$IMR = \frac{\text{deaths under age 1}}{\text{live births in the year}} \times 1000$$

RATE When a rate is mostly noise

Sketch the shape, and label the axes

PROBABILITY A rate is not a probability

$$q = \frac{e}{P_0}$$

PROBABILITY Prevalence and incidence

YOUR TURN Find what is wrong

THE ANSWER What was wrong, and the version to publish

Person-Years Lived

EXPOSURE What person-years are

Sketch the shape, and label the axes

EXPOSURE Reading a Lexis diagram

Sketch the shape, and label the axes

EXPOSURE Counting them, one village

EXPOSURE The mid-year population

$$\bar{P} = \frac{P_0 + P_1}{2}$$

YOUR TURN Three lives, one year

THE ANSWER Two and a half against three

CLOSING What you can do now